

Reliability-Weighted Cross-Corpus Pseudo-Supervision for Multimodal Depression and Parkinson Recognition under Partial Labels

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Abstract

Depression and Parkinson recognition from behavioral media is complicated by disjoint corpora: a depression example is not annotated for Parkinson, and a Parkinson example is not annotated for depression. We therefore formulate two independent binary tasks with masked unknown labels, rather than a mutually exclusive disease classifier. We study a frozen audio–video recipe that combines observed-label supervision, reliability-weighted direct pseudo-supervision for accepted missing targets, modality auxiliary losses, cross-modal agreement, and reliability-aware smooth Tchebycheff (RA-STCH) scalarization. The programme evaluates a temporal-audio baseline, a full task-conditioned R4 model, and an equal-parameter shared-fusion model over five frozen seeds. The shared model is the primary parsimonious paper candidate, selected before final Test reporting by a DEV-only parsimony decision; this role is not a statistical-superiority claim. Shared fusion obtains five-seed DEV Mean_Score 0.790914, full R4 0.790057, and temporal audio 0.765145. In the one-shot final evaluation, shared fusion has the highest Mean_Score point estimate on TEST_NONE, TEST_SOFT, and TEST_HARD (0.806003, 0.815349, and 0.826483), yet every predeclared paired final-Test Mean confidence interval includes zero. Controlled ablations support direct pseudo-supervision, uncertainty/reliability handling, the online audio input branch, and a narrow sample-specific pseudo-alignment control; several proposed components, including task-aware fusion and RA-STCH over simpler balancing, are not supported. The evidence concerns observed-task generalization and does not verify missing-label correctness, comorbidity recovery, clinical validity, or final-Test superiority.

1 Introduction

Depression and Parkinson recognition from speech and video may benefit from complementary behavioral cues, but practical data collection often produces fragmented supervision. In WSM, depression and Parkinson annotations arise from different corpora. A depression-corpus segment has an observed depression target but no Parkinson ground truth; the converse holds in the Parkinson corpus. Treating the unobserved task as a negative label would create supervision that was never collected. Treat-

ing all data as a healthy/depression/Parkinson softmax would additionally rule out simultaneous positive predictions. We instead preserve two independent sigmoid outputs and an observed-label mask.

This setting creates a useful but constrained transfer problem. It asks whether a model can share audio–video structure across corpora without treating corpus membership as disease truth. It does *not* permit a claim that an accepted pseudo target is the missing clinical diagnosis, that comorbidity has been recovered, or that the model is ready for clinical deployment. Our paper is consequently organized as a technical account of a completed, frozen evidence programme rather than as an open-ended optimization study.

The final model family uses an immutable TRAIN-only pseudo-target cache for accepted missing entries, reliability-weighted direct pseudo binary cross entropy (BCE), modality-specific auxiliary heads, and an agreement term. A task scalarizer combines depression and Parkinson objectives. These components describe the recipe actually used in the frozen final configurations; a component being present in the recipe is distinct from an ablation-supported claim that it is necessary. This distinction is material here: direct pseudo supervision and graded reliability are supported under their frozen ablation gates, while the tested RA-STCH contribution and advantage over simpler balancing are not supported.

The final comparison was deliberately locked before Test reporting. It contains a frozen temporal-audio baseline, full R4 trial012 as a secondary task-conditioned multimodal reference, and an equal-parameter shared-fusion model as the primary parsimonious paper candidate. The shared role was selected from DEV-only parsimony evidence, not from a final-Test ranking. Five seeds, selected checkpoints, an evaluator, paired summary procedures, and the final protocol views were frozen before the one-shot final Test invocation. Thus, Test results describe generalization of the frozen set; they do not select an epoch, threshold, architecture, role, or follow-up.

This work makes four bounded contributions:

1. It specifies an availability-aware, two-head audio–video model and the masked partial-label objective used in the frozen shared and full-R4 recipes.
2. It documents an immutable pseudo-supervision contract, including acceptance counts, detached reliability, and the precedence of observed labels.
3. It reports five-seed DEV and one-shot final-Test evi-

dence for three frozen roles, including task-wise trade-offs and paired descriptive intervals.

4. It preserves positive and negative Stage-6 mechanism evidence instead of presenting all retained training components as validated innovations.

The result is intentionally conservative. Shared fusion has the highest final-Test Mean point estimate across the three reporting views, but all predeclared paired Test Mean intervals include zero. We therefore report no Final-Test statistical-superiority claim.

2 Related Work

Video behavioral recognition. DEPART is a relevant visual reference because it studies interpretable depression and Parkinson detection from in-the-wild video using body-centric visual representations, temporal modelling, and a prototype-aware variant [7]. WSM retains two bounded DEPART-like video families as internal contextual baselines. The comparison boundary is important: DEPART’s published target/evaluation formulation differs from WSM’s two independently masked disease heads, and its reported UAR is not WSM Mean_Score. We therefore use published DEPART only as context, not as a leaderboard comparator.

Cross-modal learning under fragmented supervision. TACME studies task-aware cross-modal experts for behavioral tasks drawn from disjoint corpora [8]. The supplied manuscript motivates examining task-specific and shared multimodal interactions when labels are fragmented across datasets. In WSM, however, corpus membership is structurally coupled to which task is observed. We treat task-conditioned fusion and corpus diagnostics as falsifiable design hypotheses rather than as evidence that corpus identity causes a disease prediction.

Pseudo labels, calibration, and partial labels. Pseudo-label methods can transfer information from a teacher to unlabelled examples, but their targets are not ground truth [4]. Partial-label or missing-label supervision requires retaining the distinction between unobserved and negative labels [3]. Our cache is therefore immutable and restricted to TRAIN; observed truth overrides it, and pseudo targets/reliability are detached in the loss. The design supports controlled cross-corpus supervision but not verified recovery of unknown labels.

Multimodal representations and task balancing. The feature extractors are frozen WavLM-style audio and CLIP-family video representations [1, 6]. Smooth Tchebycheff scalarization provides a differentiable way to aggregate competing objectives [5]; gradient-conflict methods such as PCGrad provide related multi-objective context [9]. The final configurations retain RA-STCH, but the corresponding fixed-composition ablation did not

Table 1: Canonical membership and observed-task coverage. TEST_SOFT/HARD are fixed filtered reporting views; task-specific counts for them are not introduced here.

Split/view	Rows	D observed	P observed
TRAIN	6,325	3,660	2,665
DEV	933	621	312
TEST_NONE	1,364	827	537
TEST_SOFT	1,208	–	–
TEST_HARD	1,014	–	–

support either a contribution over equal weighting or an advantage over simpler balancing. The paper therefore distinguishes the implemented recipe from the supported mechanism claims.

Transcript context. A frozen XLM-R transcript model [2] supplies a standalone three-seed text baseline. Its transcript audit did not establish deterministic segment-level text alignment, so its unit is a video-level transcript rather than an invented segment transcript. This evidence is contextual only and does not establish that text adds to audio–video fusion.

3 Data and Evaluation Protocol

For canonical segment i , the target vector is $\mathbf{y}_i = (y_i^D, y_i^P)$ and the observed mask is $\mathbf{m}_i = (m_i^D, m_i^P)$. A depression-corpus row has $m_i^D = 1, m_i^P = 0$; a Parkinson-corpus row has $m_i^D = 0, m_i^P = 1$. Unknown is a missing annotation, not a negative target. Every supervised loss and metric applies the mask, and observed ground truth always overrides any pseudo field.

Split membership is fixed at the video level and is shared across streams. The canonical TRAIN set contains 6,325 segment rows: 3,660 with an observed depression label and 2,665 with an observed Parkinson label. DEV contains 933 rows (621 depression-observed and 312 Parkinson-observed), while TEST_NONE contains 1,364 rows (827 and 537, respectively). TEST_SOFT and TEST_HARD contain 1,208 and 1,014 rows. These latter views are filtered reporting protocols, not development sets or a model-selection hierarchy.

For each task, the fixed metric is

$$\begin{aligned} \text{Score}_t &= \frac{\text{UAR}_t + \text{MF1}_t}{2}, \\ \text{Mean_Score} &= \frac{\text{Score}_D + \text{Score}_P}{2}. \end{aligned} \tag{1}$$

DEV Mean_Score was the sole signal for checkpoint selection. The final reports use segment-level metrics as primary outputs. The predeclared five-seed summaries use arithmetic means, sample standard deviations, and descriptive two-sided Student- t intervals (multiplier 2.7764451051977987); they are not multiplicity-corrected tests or a license to infer superiority.

3.1 Immutable pseudo-target contract

The final training cache is the immutable, TRAIN-only `train_missing_targets.pt` file:

```
/media/maxim/Programs/Features/WSM/ramps_r2_semantic_v1/train_
missing_targets.pt
SHA256:17cf5e67e8c244d81b7c21f0842988966a23d83d69e296f7c5a5181376d6b945
```

Of the 6,325 TRAIN rows, 2,665 depression and 3,660 Parkinson targets are missing by the canonical mask. The cache accepts 376 depression entries (376 positive, 0 negative) and 1,801 Parkinson entries (212 positive, 1,589 negative). Each accepted target equals the calibrated probability from the frozen audio teacher at that entry. Reliability lies in $[0, 1]$ and is detached; rejected and observed pseudo fields are neutral. Accepted entries cannot overlap observed truth, and Test receives no pseudo supervision.

Semantic and video evidence participated in the frozen acceptance/reliability construction where applicable, but the exact Stage-6 conclusion is *SEMANTIC-EVIDENCE CONTRIBUTION VIA DEPRESSION PSEUDO ACCEPTANCE NOT SUPPORTED*. This evidence neither makes semantic information an established necessity nor permits a missing-label-correctness claim.

4 Model and Training Objective

4.1 Availability-aware shared fusion

The primary model is the frozen disease-query architecture with `task_aware_fusion=false`. Let x_a be the online audio class representation, and let X_v and m_v be temporal video features and their valid-position mask. With projections P_a, P_v and masked temporal pooling,

$$h_a = P_a(x_a), \quad h_v = P_v\left(\frac{\sum_j m_{vj} X_{vj}}{\max(1, \sum_j m_{vj})}\right). \quad (2)$$

An unavailable modality is zeroed before projection and its projected feature is also masked after projection. Consequently the corresponding gate weight is zero and no unavailable modality feature is introduced as a fabricated signal.

The implementation begins with two learned disease queries q_D, q_P , but the primary model uses their mean:

$$\bar{q} = \frac{1}{2}(q_D + q_P). \quad (3)$$

Let LN_t be the two candidate normalizations. Its shared modality candidates are

$$\begin{aligned} c_a &= \frac{1}{2} \sum_{t \in \{D, P\}} \text{LN}_t(h_a + \bar{q}), \\ c_v &= \frac{1}{2} \sum_{t \in \{D, P\}} \text{LN}_t(h_v + \bar{q}). \end{aligned} \quad (4)$$

A shared gate g scores the query-candidate concatenations:

$$s_a = g([\bar{q}; c_a]), \quad s_v = g([\bar{q}; c_v]). \quad (5)$$

For availability vector $\mathbf{a} = (a_a, a_v)$, unavailable scores are set to $-\infty$ before the softmax:

$$(\alpha_a, \alpha_v) = \text{softmax}_{\mathbf{a}}(s_a, s_v). \quad (6)$$

The fused feature first combines candidates and then averages the two fusion normalizations,

$$\begin{aligned} u &= \bar{q} + \alpha_a c_a + \alpha_v c_v, \\ h &= \frac{1}{2} \sum_{t \in \{D, P\}} \text{LN}_t^f(u), \quad z_t = f_t(h). \end{aligned} \quad (7)$$

The two output heads f_D, f_P remain independent sigmoid-logit heads. Full R4 instead retains a separate $q_t, c_{a,t}, c_{v,t}$, gate, and fused representation for each task. Both models have exactly 295,239 trainable downstream parameters; the shared model is a simplification in conditioning, not a reduction in model size.

4.2 Masked multi-component objective

For task t , observed mask m_{it} , accepted pseudo mask a_{it} , logits z_{it} , and BCE-with-logits ℓ , the observed component is

$$L_{\text{obs},t} = \frac{\sum_i m_{it} \ell(z_{it}, y_{it})}{\sum_i m_{it} + \epsilon}. \quad (8)$$

For missing accepted entries, $\tilde{y}_{it}$ and r_{it} are detached pseudo target and reliability:

$$L_{\text{pseudo},t} = \frac{\sum_i (1 - m_{it}) a_{it} r_{it} \ell(z_{it}, \tilde{y}_{it})}{\sum_i (1 - m_{it}) a_{it} r_{it} + \epsilon}. \quad (9)$$

This operation does not turn unaccepted entries into negatives. The cache invariant $a_{it} m_{it} = 0$ makes observed-label precedence explicit.

The architecture also emits audio and video auxiliary logits z_{it}^a, z_{it}^v and availability indicators b_i^a, b_i^v . Auxiliary observed-label BCE is

$$L_{\text{aux},t} = \frac{\sum_i m_{it} [b_i^a \ell(z_{it}^a, y_{it}) + b_i^v \ell(z_{it}^v, y_{it})]}{\sum_i m_{it} (b_i^a + b_i^v) + \epsilon}, \quad (10)$$

and agreement applies only where the task is observed and both modalities are available:

$$L_{\text{agree},t} = \frac{\sum_i m_{it} b_i^a b_i^v [\sigma(z_{it}^a) - \sigma(z_{it}^v)]^2}{\sum_i m_{it} b_i^a b_i^v + \epsilon}. \quad (11)$$

The per-task objective used by the frozen final configs is

$$L_t = L_{\text{obs},t} + \lambda_p(e) L_{\text{pseudo},t} + \lambda_{\text{aux}} L_{\text{aux},t} + \lambda_{\text{agree}} L_{\text{agree},t}. \quad (12)$$

The pseudo schedule has three observed-only epochs, a five-epoch ramp, and final $\lambda_p = 1.0$. The selected constants are $\lambda_{\text{aux}} = 0.4121598308$, $\lambda_{\text{agree}} = 0.4812668453$, and $\tau = 0.04871016345$.

4.3 RA-STCH in the frozen recipe

With detached controller weights w_t , the active two-task scalarizer is

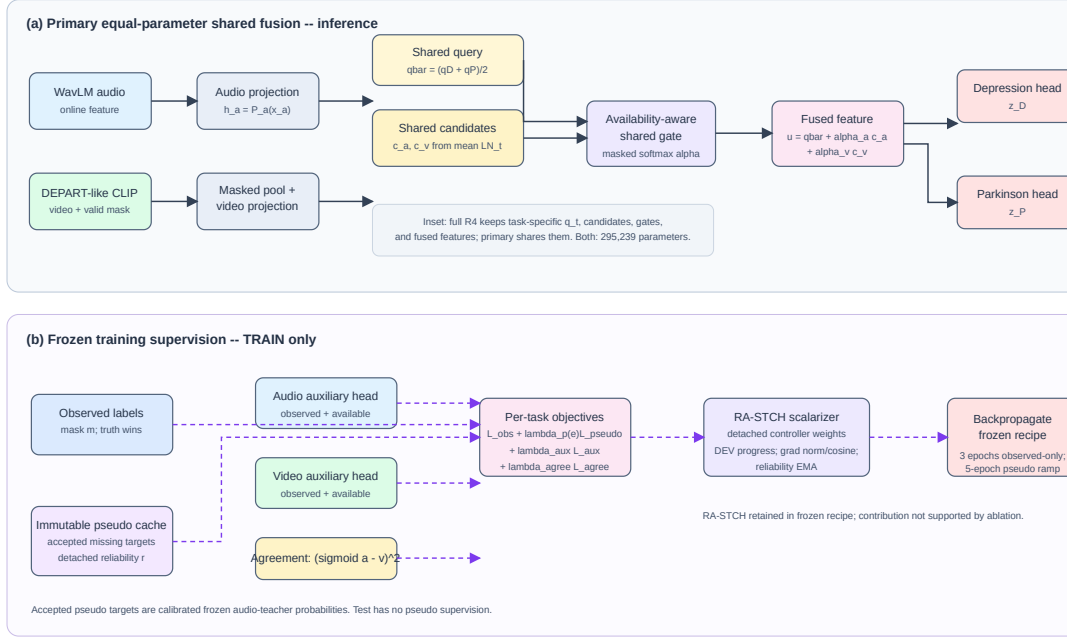
$$L = \tau \log \sum_{t \in \{D, P\}} \exp(w_t L_t / \tau). \quad (13)$$

The RA-STCH controller updates bounded weights using DEV progress relative to frozen references, gradient-norm and gradient-cosine diagnostics, and a pseudo-reliability exponential moving average. This is a description of the trained configuration, not a validated causal explanation. The frozen ablation conclusions are *RA-STCH BALANCING CONTRIBUTION OVER EQUAL NOT SUPPORTED* and *RA-STCH ADVANTAGE OVER SIMPLER BALANCING NOT SUPPORTED*. Thus RA-STCH remains reported for reproducibility while no necessity or general advantage claim is made.

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WSM: availability-aware shared fusion and frozen training recipe

Solid arrows: inference. Dashed violet arrows: training-only supervision. Two independent sigmoid-logit outputs.



file:///media/maxim/Programs/Projects/WSM/paper/figures/wsm_method_overview.svg

1/1

Figure 1: Two-panel vector overview of the primary equal-parameter shared-fusion model and its frozen training recipe. Solid arrows denote inference; dashed arrows are training-only supervision paths. Full R4 changes shared query/candidate/gating/fusion to task-specific versions but has the same trainable parameter count (295,239). RA-STCH is retained in the frozen recipe; its contribution is not supported by the Stage-6 ablation.

5 Frozen Experimental Design

The final compared set was locked before Test reporting: temporal audio, full R4 trial012, and equal-parameter shared fusion, each with seeds 42–46. For every run, the selected checkpoint is the epoch with maximum DEV Mean_Score; the selected checkpoint identity was then frozen before the final evaluator read any same-seed Test outputs. The one-shot evaluator consumed the 15 predeclared checkpoint/config identities under the frozen reporting contract. No threshold fitting, recalibration, retry, architecture modification, or role revision followed final Test reporting.

Frozen roles. Shared fusion is the primary parsimonious paper candidate. Full R4 trial012 is the secondary multimodal reference, and frozen temporal audio is the baseline. This assignment was a DEV-only parsimony decision: shared and full R4 have equal parameter count and their paired DEV Mean interval did not separate them. It is not a Final-Test ranking or superiority statement.

Comparisons and summaries. Primary metrics are segment-level disease Scores and Mean_Score. The predeclared same-seed paired summaries are full R4–audio, shared–audio, and shared–R4. We show their frozen means and descriptive 95% intervals, but do not reinterpret them as multiplicity-corrected statistical tests. Test views are TEST_NONE, TEST_SOFT, and TEST_HARD; they are all reports of the same frozen model set.

Calibration diagnostic. Calibration is DEV-only and diagnostic. It uses observed rows only (621 depression and 312 Parkinson rows per seed), sigmoid probabilities, 15 equal-width bins, Brier score, and ECE-15. No recalibration, threshold search, or class balancing is applied. Table 2 copies the accepted five-seed summaries rather than re-estimating any value.

Text and description context. T1 is a completed standalone transcript baseline using frozen XLM-R features. Its three-seed DEV D/P/Mean is 0.6268063333/0.8447246667/0.7357653333, with visibly weaker and more seed-sensitive depression performance. It is not evidence that text improves A+V fusion. The planned T2 transcript-plus-observable-description family stopped before cache construction, model implementation, or training: its frozen preflight found demographic/identity inference in 4/24 outputs and judged only 18/24 outputs concise enough. T2 therefore has no performance claim.

Table 2: Accepted five-seed DEV calibration means. Counts are observed DEV rows; values are diagnostics only.

Method	Task	Brier	ECE-15
Audio	D	.217045	.163912
Audio	P	.145966	.132109
Full R4	D	.183842	.085347
Full R4	P	.087923	.078154
Shared	D	.179846	.084853
Shared	P	.088764	.084624

6 Results and Discussion

6.1 Five-seed DEV evidence

Table 3 gives the frozen primary comparison. Both multimodal finalists exceed temporal audio Mean_Score on all five DEV seeds. The paired DEV Mean difference for full R4–audio is +0.024912 ([+0.001169, +0.048655]), and for shared–audio it is +0.025769 ([+0.009556, +0.041984]). The shared–R4 difference is +0.000858 ([−0.009704, +0.011419]), so the two multimodal finalists are not separated by the predeclared paired DEV interval. This is the DEV-only basis for retaining the equal-parameter shared model as the parsimonious primary candidate and full R4 as the secondary multimodal reference.

6.2 One-shot final-Test reporting

Shared fusion has the highest five-seed Mean point estimate on TEST_NONE, TEST_SOFT, and TEST_HARD. This pattern is descriptive, not a final-Test superiority result. Every predeclared paired Test Mean interval contains zero (Table 5), including shared–audio and shared–R4 on every view. The Test outcome therefore neither changes frozen roles nor selects a model, architecture, threshold, or follow-up.

6.3 Task trade-off rather than a single undifferentiated gain

The aggregate Mean can conceal a disease-specific trade-off. On TEST_NONE, shared fusion raises depression Score from .752954 (audio) to .799233, a +.046279 difference, while Parkinson Score falls from .848230 to .812774, a −.035456 difference; the resulting Mean difference is only +.005411. The same broad pattern appears in the final summaries: audio is particularly strong for Parkinson, whereas A+V fusion improves depression more consistently and can give back Parkinson performance. This is why D/P/Mean are reported together and why a small aggregate Test difference is not used to revise the frozen roles.

Table 3: Frozen five-seed main comparison. DEV cells are D/P/Mean; final-Test cells are Mean with predeclared descriptive 95% intervals. Shared fusion’s primary role was fixed before Test reporting.

Method	Modality	Seeds	DEV D/P/Mean	TEST_NONE Mean	TEST_SOFT Mean	TEST_HARD Mean
Temporal audio	A	42–46	.721720/.808569/.765145	.800592 [.781012,.820172]	.813068 [.793383,.832754]	.824268 [.808625,.839910]
Full R4 trial012	A+V	42–46	.731250/.848863/.790057	.803502 [.780018,.826987]	.811343 [.785432,.837253]	.821767 [.790908,.852626]
Shared fusion	A+V	42–46	.737690/.844139/.790914	.806003 [.794417,.817590]	.815349 [.801248,.829450]	.826483 [.810207,.842758]

Table 4: Final-Test disease-score breakdown (D/P/Mean). These are frozen monitoring/reporting summaries, not additional selection results.

Method	Protocol	Depression Score	Parkinson Score	Mean_Score
Temporal audio	TEST_NONE	.752954	.848230	.800592
Full R4 trial012	TEST_NONE	.790977	.816028	.803502
Shared fusion	TEST_NONE	.799233	.812774	.806003
Temporal audio	TEST_SOFT	.775925	.850212	.813068
Full R4 trial012	TEST_SOFT	.802296	.820389	.811343
Shared fusion	TEST_SOFT	.812757	.817941	.815349
Temporal audio	TEST_HARD	.788289	.860246	.824268
Full R4 trial012	TEST_HARD	.806044	.837490	.821767
Shared fusion	TEST_HARD	.815916	.837049	.826483

6.4 Internal video evidence and external DEPART context

The two internal DEPART-like video models exhibit a related task trade-off (Table 6). V2 was selected over V1 by the predeclared DEV Mean rule: V2–V1 is $-.041822$ for depression, $+.048417$ for Parkinson, and $+.003298$ for Mean. At the same DEV-selected epoch, V2’s Test Mean monitoring values are lower than V1’s on NONE, SOFT, and HARD. Those Test values did not drive family selection and are not a retrospective reversal of the DEV-only decision.

Published DEPART remains contextual rather than directly matched (Table 7). It uses a different target/evaluation formulation and UAR, and its manually cleaned Test must not be equated with WSM TEST_HARD. The published results are useful for situating strong visual depression recognition, but not for declaring a WSM leaderboard outcome.

6.5 Stage-6 mechanism record

Table 8 separates the frozen quantitative comparisons from their narrow conclusions. Direct pseudo-supervision and uncertainty/reliability have supported claims under their exact gates, although the latter effect is small. The no-audio result supports the online-audio-input contribution under the unchanged pseudo-training pipeline. By contrast, no-video fails its predeclared contribution criterion despite a lower aggregate ablation Mean, because the disease-specific condition fails. The model still contains RA-STCH, auxiliary heads, and agreement in the frozen final recipe; their inclusion should not be read as an unqualified necessity claim.

The corpus diagnostic did not trigger the strong linear-

decodability flag, and fusion amplification of corpus decodability was not supported. Neither outcome licenses a causal corpus-shortcut claim. T1 provides a different context: it has strong Parkinson but weak, seed-sensitive depression performance and is a standalone text baseline, not additive A+V evidence. T2 was blocked at preflight and has no performance record.

Table 5: Frozen paired final-Test Mean deltas (mean [95% CI]). Every interval includes zero.

Protocol	R4 – audio	shared – audio	shared – R4
TEST_NONE	+.002910 [−.021829, .027649]	+.005411 [−.014285, .025108]	+.002501 [−.013563, .018565]
TEST_SOFT	−.001726 [−.029386, .025934]	+.002281 [−.014017, .018578]	+.004006 [−.014503, .022515]
TEST_HARD	−.002501 [−.029933, .024931]	+.002215 [−.013123, .017553]	+.004716 [−.015706, .025137]

Table 6: Internal DEPART-like video baselines. V2 was selected by DEV Mean before its monitoring Test values were considered.

Model	DEV D	DEV P	DEV Mean	TEST_NONE Mean	TEST_SOFT Mean	TEST_HARD Mean
DEPART-like V1	.661923	.744626	.703274	.732454	.742796	.751864
DEPART-like V2	.620101	.793043	.706572	.722250	.721935	.730925

Table 7: Published DEPART context (UAR only; different target/evaluation formulation; cleaned DEPART Test $\neq$ WSM TEST_HARD unless independently established; not a direct leaderboard comparison).

Published setting	D UAR	P UAR
Multitask, original Test	82.25%	66.82%
Multitask, manually cleaned Test	83.67%	80.71%
Best single task, original Test	85.55%	78.58%
Best single task, manually cleaned Test	88.08%	77.96%

Table 8: Quantitative companion to the frozen Stage-6 claim ledger. Not-supported results are scoped to the stated ablation/comparator, not a universal uselessness finding.

Component/control	Frozen comparison	Exact interpretation
Direct pseudo BCE	No-direct Mean .769878; R4 .7869088179; $\Delta = +.0170308179$; full wins 2/3	DIRECT PSEUDO-SUPERVISION CONTRIBUTION SUPPORTED. Only the direct pseudo BCE term is tested.
Graded reliability	Uniform Mean .784922; R4-uniform $+.0019868179$	UNCERTAINTY/RELIABILITY CONTRIBUTION SUPPORTED; aggregate effect is small.
No video	No-video Mean .766431; full-no-video Mean $+.020478$, D $-.014389$	VIDEO MODALITY CONTRIBUTION NOT SUPPORTED under its exact criterion.
No online audio	No-audio Mean .722703; full-no-audio Mean $+.064206$; full wins 3/3	ONLINE AUDIO INPUT MODALITY CONTRIBUTION SUPPORTED. Historical audio teacher information remains in the cache.
Shared fusion	Shared Mean .7935663333; full-shared $-.0066575154$; full wins 0/3	TASK-AWARE FUSION CONTRIBUTION NOT SUPPORTED.
Sparse-MTL isolation	Isolated Mean .799671; full-isolated $-.012762$	SPARSE MTL JOINT-TRAINING CONTRIBUTION NOT SUPPORTED; comparator uses two separately trained models.
RA-STCH audit	Equal .7791726667; static .7803973333; progress .784751; RA .7725573333	RA-STCH BALANCING CONTRIBUTION OVER EQUAL NOT SUPPORTED; RA-STCH ADVANTAGE OVER SIMPLER BALANCING NOT SUPPORTED.
No semantic D acceptance	Mean .791408; full-minus D/P/Mean $+.020809$ / $-.029808$ / $-.004499$	SEMANTIC-EVIDENCE CONTRIBUTION VIA DEPRESSION PSEUDO ACCEPTANCE NOT SUPPORTED.
Shuffled pseudo control	Shuffled .737118; matched .7791726667; matched-minus $+.0420546667$; matched wins 3/3	SAMPLE-SPECIFIC PSEUDO ALIGNMENT CLAIM SUPPORTED BY THIS NEGATIVE CONTROL; no correctness or comorbidity claim follows.

7 Limitations and Scope

Partial labels are the central limitation, not a cosmetic training detail. The observed mask prevents unknown disease labels from becoming negatives, but it does not supply a dual-annotated cohort. We therefore do not claim verified correctness of missing labels, recovery of true comorbidity, or clinical correctness of pseudo probabilities. Direct pseudo-supervision evidence concerns observed-task generalization under the frozen data contract.

The primary shared-fusion role is a frozen DEV-only parsimony decision. It is not a claim that shared fusion is statistically superior to full R4 or audio on final Test. Every predeclared paired final-Test Mean interval includes zero, and no post-Test threshold, model, checkpoint, or role revision is permitted. Five seeds improve descriptive stability relative to a single run but do not justify broad scientific or clinical generalization beyond the stated protocol.

The no-audio ablation removes the online student audio feature branch, not all historical audio-derived information: the immutable pseudo cache stores calibrated frozen audio teacher probabilities. Conversely, the no-video ablation preserves architecture and masks online video availability. These are precisely scoped branch-removal experiments, not universal statements about modality value. Likewise, unsupported component findings do not show that those components are universally useless.

Text and description evidence is intentionally limited. T1 is a standalone video-level-transcript baseline; it does not demonstrate an additive gain when combined with A+V. T2 is blocked before cache, model, training, or evaluation, so it cannot support a performance claim. DEPART figures are external context with a different target/evaluation formulation, and its manually cleaned Test is not WSM TEST_HARD.

Finally, corpus identity remains structurally coupled to observed-task identity. The corpus probe did not trigger its strong-decodability flag, but this does not prove the absence of corpus effects or establish causality. No further experiment, Test invocation, or post hoc mechanism search is authorized by the completed programme.

8 Conclusion

We present a frozen evidence record for multimodal depression and Parkinson recognition under disjoint-corpus partial labels. The method preserves two independent disease logits, masks unknown labels, uses immutable TRAIN-only reliability-weighted pseudo targets only where accepted, and combines online audio/video features with availability-aware shared fusion. The full task-conditioned R4 configuration remains a secondary reference; equal-parameter shared fusion is the primary parsimonious paper candidate by a DEV-only pre-Test decision; temporal audio remains the baseline.

Five-seed DEV evidence supports a multimodal Mean advantage over audio and does not separate shared fusion from full R4 under the frozen paired interval. In the one-shot final Test report, shared fusion has the highest Mean point estimate across all three views, but all predeclared paired intervals include zero. We therefore make no Final-Test superiority claim. The ablation record supports direct pseudo-supervision, uncertainty/reliability handling, the online audio input branch, and a narrow sample-specific pseudo-alignment control; it preserves negative findings for several other mechanisms. The resulting claims concern observed-task generalization only, not clinical diagnosis, missing-label correctness, or comorbidity recovery.

Table 9: Final architecture identities. Both multimodal variants have 295,239 trainable parameters.

Item	Frozen identity
Primary shared fusion	<code>wsm_av_r3_disease_query_model;</code> <code>task_aware_fusion=false;</code> audio/video dimensions 768/512; hidden/gate dimensions 160/96; dropout .10034630487649018.
Secondary full R4	Same model family and dimensions; task-aware fusion enabled; separate task-conditioned query/candidate/gating/fusion path.
Output contract	Two independent depression/Parkinson sigmoid logits; unknown targets masked rather than converted to negative.

Table 10: Frozen optimization recipe for the final multimodal configurations. RA-STCH is reported as implemented, not as a supported superiority mechanism.

Component	Frozen value
Optimizer	AdamW; learning rate .000021217831107180076; weight decay .0001855295766514827.
Training	30 epochs; gradient clip 0.5; mixed precision enabled.
Scalarization	<code>ra_stch</code> ; $\tau = .0487101634503887$.
Loss weights	Auxiliary .41215983081110485; agreement .48126684525457264.
Pseudo schedule	3 observed-only epochs; 5 ramp epochs; final pseudo scale 1.0.

A Reproducibility and Evidence Traceability

This appendix records the frozen identities and accepted evidence used by the manuscript. It is a traceability supplement, not a new experiment or analysis. The final roles remain unchanged: equal-parameter shared fusion is the primary parsimonious candidate by the pre-Test DEV-only decision, full R4 trial012 is the secondary multimodal reference, and temporal audio is the baseline. The one-shot Final Test did not revise these roles.

A.1 Final model and optimization identities

Cache SHA256: 17cf5e67e8c244d81b7c21f0842989866a23d83d69e296f7c5a5181376d6b945.

Accepted pseudo targets equal calibrated frozen audio-teacher probabilities at accepted missing entries. Reliability is detached; observed truth overrides pseudo supervision; rejected and observed pseudo fields are neutral; and no Test pseudo supervision exists.

A.2 Final checkpoint and evaluator protocol

The final seed set is 42, 43, 44, 45, and 46 for temporal audio, full R4 trial012, and shared fusion. All 15 checkpoint identities were frozen before Final Test; their paths, SHA256 values, selected epochs, and run identities are listed in the accepted Stage-7 DEV evidence dossier. The standardized Final Test ran exactly once after the evaluator firewall.

A.3 Final five-seed values

Table 13 copies accepted selected-checkpoint DEV Scores (not UAR or macro-F1). Table 14 copies accepted Final-Test Mean Scores. These rows are transcribed from the Stage-7 DEV/Test dossiers; no metric is recomputed here.

Table 11: Immutable TRAIN-only pseudo-cache contract.

Quantity	Frozen value
TRAIN rows	6,325
Missing depression/Parkinson	2,665 / 3,660
Accepted depression/Parkinson	376 / 1,801
Accepted positive D/P	376 / 212
Accepted negative D/P	0 / 1,589

A.4 Stage-6 claim matrix

Table 15 maps each frozen component interpretation to the accepted evidence task. Supported and not-supported outcomes remain narrow to their stated controls; they are neither significance claims nor general statements of universal usefulness.

The corpus findings do not support a causal corpus-shortcut claim. No claim-matrix row establishes pseudo-label correctness, missing-label recovery, comorbidity recovery, clinical validity, final-method promotion, or Final-Test superiority.

Table 12: Final evaluator and reporting identity. The report is a frozen one-shot Final-Test artifact, not a model-selection input.

Item	Frozen identity
Evaluator firewall commit	bfaae7f812904dd9ae9a7460b394117867db5439
Evaluator SHA256	b691c9563bb6ffd63316b664949dcf1bef02d8d3f4e770110c2e70adbc4cee1e
Final report	/media/maxim/Programs/Features/WSM/stage7_final_test/final_test_v1.json
Report SHA256	59f763f04843c8e54d5d3f9893d3abe0530b47a32ea11f5b9585390edeb515bc
Protocol rows	TEST_NONE 1,364; TEST_SOFT 1,208; TEST_HARD 1,014.

Table 13: Accepted per-seed DEV D/P/Mean Scores for the final frozen methods.

Method	Seed	Depression	Parkinson	Mean
Temporal audio	42	.747918	.827735	.787827
Temporal audio	43	.708699	.811771	.760235
Temporal audio	44	.734396	.809382	.771889
Temporal audio	45	.697398	.799034	.748216
Temporal audio	46	.720187	.794924	.757556
Full R4 trial012	42	.759025	.879259	.819142
Full R4 trial012	43	.718527	.804363	.761445
Full R4 trial012	44	.728217	.832060	.780139
Full R4 trial012	45	.725077	.858468	.791772
Full R4 trial012	46	.725405	.870164	.797785
Shared fusion	42	.765324	.883606	.824465
Shared fusion	43	.724496	.818152	.771324
Shared fusion	44	.731448	.838372	.784910
Shared fusion	45	.739733	.834672	.787202
Shared fusion	46	.727451	.845891	.786671

Table 14: Accepted per-seed Final-Test Mean_Scores. The three protocol views are reporting outputs only and did not select or revise any role.

Method	Seed	TEST_NONE	TEST_SOFT	TEST_HARD
Temporal audio	42	.809486	.815156	.828135
Temporal audio	43	.791690	.802602	.818189
Temporal audio	44	.811919	.820354	.831768
Temporal audio	45	.776989	.793148	.805612
Temporal audio	46	.812876	.834083	.837633
Full R4 trial012	42	.801561	.811496	.809755
Full R4 trial012	43	.822225	.828142	.841820
Full R4 trial012	44	.822847	.833456	.851243
Full R4 trial012	45	.780446	.781625	.789749
Full R4 trial012	46	.790433	.801993	.816267
Shared fusion	42	.801124	.811002	.815546
Shared fusion	43	.819130	.826236	.839551
Shared fusion	44	.808215	.818077	.831848
Shared fusion	45	.793969	.797907	.809696
Shared fusion	46	.807580	.823524	.835773

Table 15: Claim traceability matrix. Conclusions reproduce the frozen Stage-6 wording.

Hypothesis/component	Exact frozen conclusion	Status	Evidence task
Sparse MTL	SPARSE MTL JOINT-TRAINING CONTRIBUTION NOT SUPPORTED	Not supported	TASK-006J
Task-aware fusion	TASK-AWARE FUSION CONTRIBUTION NOT SUPPORTED	Not supported	TASK-006I
Direct pseudo supervision	DIRECT PSEUDO-SUPERVISION CONTRIBUTION SUPPORTED	Supported	TASK-006D
Uncertainty/reliability	UNCERTAINTY/RELIABILITY CONTRIBUTION SUPPORTED	Supported	TASK-006E
Semantic depression pseudo acceptance	SEMANTIC-EVIDENCE CONTRIBUTION VIA DEPRESSION PSEUDO ACCEPTANCE NOT SUPPORTED	Not supported	TASK-006F
RA-STCH	RA-STCH BALANCING CONTRIBUTION OVER EQUAL NOT SUPPORTED; RA-STCH ADVANTAGE OVER SIMPLER BALANCING NOT SUPPORTED	Not supported	TASK-006C
Video modality	VIDEO MODALITY CONTRIBUTION NOT SUPPORTED	Not supported	TASK-006G
Online audio input modality	ONLINE AUDIO INPUT MODALITY CONTRIBUTION SUPPORTED	Supported	TASK-006H
Pseudo alignment negative control	SAMPLE-SPECIFIC PSEUDO ALIGNMENT CLAIM SUPPORTED BY THIS NEGATIVE CONTROL	Supported	TASK-006A-FIX2
Corpus probe: strong flag	STRONG CORPUS-ID DECODABILITY FLAG NOT TRIGGERED	Diagnostic	TASK-006B
Corpus fusion amplification	FUSION AMPLIFICATION OF CORPUS DECODABILITY NOT SUPPORTED	Not supported	TASK-006B
Equal-parameter control	Covered by TASK-006I	Complete	TASK-006I

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